

Outsourcing: Applications Development SLAs

As users become more comfortable with outsourcing applications development, the question of how to manage these deals looms large. We propose a set of measurements that can help users measure and manage these deals.

Core Topic

IT Management: Outsourcing —
Management Issues, Trends and Vendors

Key Issues

What are the trends in applications
outsourcing?

What are the critical success factors and
trends in outsourcing?

Definition of Selected Measures

Function points: An application size
measure based on the data accessed and
manipulated, and the processes that do so;
essential as a normalizer for productivity,
quality and cost metrics.

Cyclomatic complexity: A measure
developed by Thomas McCabe that
quantifies the complexity of logic by
examining the number of unique paths
possible through the code.

Essential complexity: A measure developed
by Thomas McCabe that examines the
number of possible entry and exit points in
a component.

The Problem

As discussed in ESPE *Research Note* KA-OUT-043, 21 June 1996, measuring various service dimensions of an applications outsourcing contract (e.g., cost, quality, service productivity, customer satisfaction and personnel) is critical to the success of the deal. However, few organizations know how to do this effectively. As a result, many of our inquiries regarding applications outsourcing focus on how organizations can measure, monitor and manage their deals.

A Proposed Solution

A more robust set of measurements is needed to help users monitor and manage their applications outsourcing deals. These measurements should be formalized in a service-level agreement (SLA) between the outsourcer and client. While many of today's outsourcing contracts include comprehensive SLAs for data center outsourcing deals or desktop outsourcing deals, the same cannot be said of SLAs in applications development contracts.

In ESPE *Research Note* KA-OUT-043, 21 June 1996, we proposed a set of metrics to be used to help manage applications maintenance outsourcing deals. In this *Research Note*, we propose a different set of measurements and sample service goals that users can employ to measure and manage their applications development outsourcing deals (see Figure 1). It is important to note that the proposed measurements may be used across organizations; however, the actual service levels will vary widely by enterprise. The sample service goals listed below should be considered as a starting point for organizations looking at this issue. Users should modify them to meet the needs of their particular organization and outsourcing arrangement.

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Figure 1
Sample Service Levels: Development and Enhancement

Function: Applications

Performance Area: Development and Enhancement

Sample Service Levels		
Service Dimension	Measurement	Sample Service Goals
1. Cost	Cost/FP by application, attribute and severity level Plan vs. actual	Cost/FP of < \$x Accuracy of x% on planned vs. actual costs
2A. Quality — Functional Do the applications meet the functional requirements?	Response times — by transaction Defect levels/FP during warrantee period, by severity	All response requirements met Defect levels of x per FP, 0 severity 1 defects For enhancements, 0 increase in residual rates
2B. Quality — Technical Does the delivered product meet the specifications?	Essential complexity Cyclomatic complexity Dead code Unstructured code Compliance with installation coding standards	<x per module <x per module 0 Dead code x% unstructured code 100% compliance, ensured through inspections
2C. Quality — Process How did the process perform during the life of the project?	Defect removal rate — overall, by phase Defect density/FP — overall, by phase introduced	Defect removal rates of >x% Defect densities of < x/FP
3. Service Productivity	Productivity/project, in FP/hour Cycle time/project, in FP/ day or days/FP Project attributes Work backlog, in FPs Productivity/phase, in FP/hour Effort, by phase Planned/actual dates, effort, FP counts	x FP/hour, depending on attributes Specific cycle time requirements x% new dev. using specific methods/tools Backlog < x months x FP/hour in analysis/design x% spent in analysis/design x% accuracy on dates, effort, FP counts
4A. Customer Satisfaction — Project-Based	Reliability: Was the project on time? Did the project meet business requirements? Staff professionalism, responsiveness and availability Documentation: Was it useful, clear and complete? Overall satisfaction	x% projects delivered on time Scale-based opinion survey Scale-based opinion survey Scale-based opinion survey Scale-based opinion survey
4B. Customer Satisfaction — Periodic	Reliability: Do products and services have high quality? Responsiveness: Does IT provide timely solutions?	Scale-based opinion survey Scale-based opinion survey
4C. Customer Satisfaction — Return	Return: Does IT provide an adequate ROI?	Scale-based opinion survey, OR activity-based
5. Personnel	Turnover of staff by skill level and position Availability of staff by skill level and position Elapsed time to get staff by skill level and position Training hours per FTE by tool and methodology	Turnover of < 20%, 100% accessibility during avail. n% project managers with > x years experience avail. Get skilled C++ programmers within 1 month 5 days C++ training for programmers with 1 year exp.

Source: Gartner Group

Acronym Key

FP	Function point
FTE	Full time equivalent
ROI	Return on investment

Bottom Line: SLAs are essential for managing applications development outsourcing deals. Users should examine the service elements listed above and adapt them to their own requirements.

